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Experiments 22 – 25

Experiment No: 22

DATABASE CONNECTIVITY USING PHP AND MYSQL

AIM

To connect PHP with MySQL and execute CREATE, INSERT, SELECT, UPDATE and DELETE commands on a database table using PHP.

PROCEDURE

- Install and start XAMPP/WAMP so that Apache and MySQL services are running.
- Place the PHP script inside the htdocs folder of the XAMPP installation.
- Establish a connection to the MySQL server from PHP using mysqli_connect().
- Create the database and the required table (emp) if they do not already exist.
- Execute an INSERT query to add a record into the emp table.
- Execute a SELECT query to retrieve and display the records.
- Execute UPDATE and DELETE queries to modify and remove records.
- Close the database connection using mysqli_close().
- Run the script through a browser at localhost and observe the output at each stage.

EXPECTED OUTPUT

Browser output confirms a successful connection to MySQL ("Connected successfully").

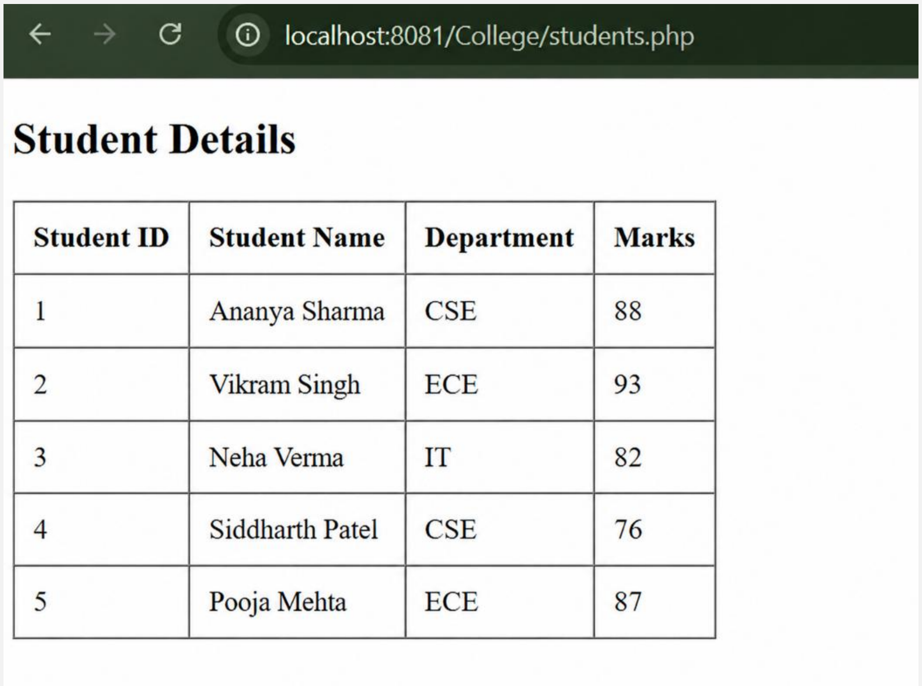
After the CREATE and INSERT operations, the emp table is created and shows the inserted record (id, name, empsalary).

The SELECT query output lists all rows currently present in the emp table.

After the UPDATE operation, the empsalary field for the given id is changed to the new value.

After the DELETE operation, the specified record no longer appears in the SELECT output.

OUTPUT SCREENSHOT:



Student Details

Student ID	Student Name	Department	Marks
1	Ananya Sharma	CSE	88
2	Vikram Singh	ECE	93
3	Neha Verma	IT	82
4	Siddharth Patel	CSE	76
5	Pooja Mehta	ECE	87

RESULT: Thus PHP was successfully connected to MySQL and the CREATE, INSERT, SELECT, UPDATE and DELETE commands were executed.

Experiment No: 23

TRAIN TICKET RESERVATION SYSTEM

AIM

To design and implement an online train ticket reservation system that allows a user to book, view and cancel a train ticket, with all passenger and booking details stored and retrieved from a database.

PROCEDURE

- Ensure the latest version of Python and the required database (SQLite/MySQL) are installed.
- Install the required modules (e.g. sqlite3, tkinter) using pip if not already available.
- Create the required database and the passengers table to hold booking details.
- Establish a connection between the Python program and the database.
- Implement the booking function to accept passenger details (name, age, gender, source, destination, date, class) and generate a unique ticket ID.
- Implement the view function to retrieve and display ticket details using the ticket ID.
- Implement the cancellation function to delete a booking record from the database.
- Assemble all functions into a menu-driven program and run it.
- Test each operation (book, view, cancel) and verify the database is updated correctly.

EXPECTED OUTPUT

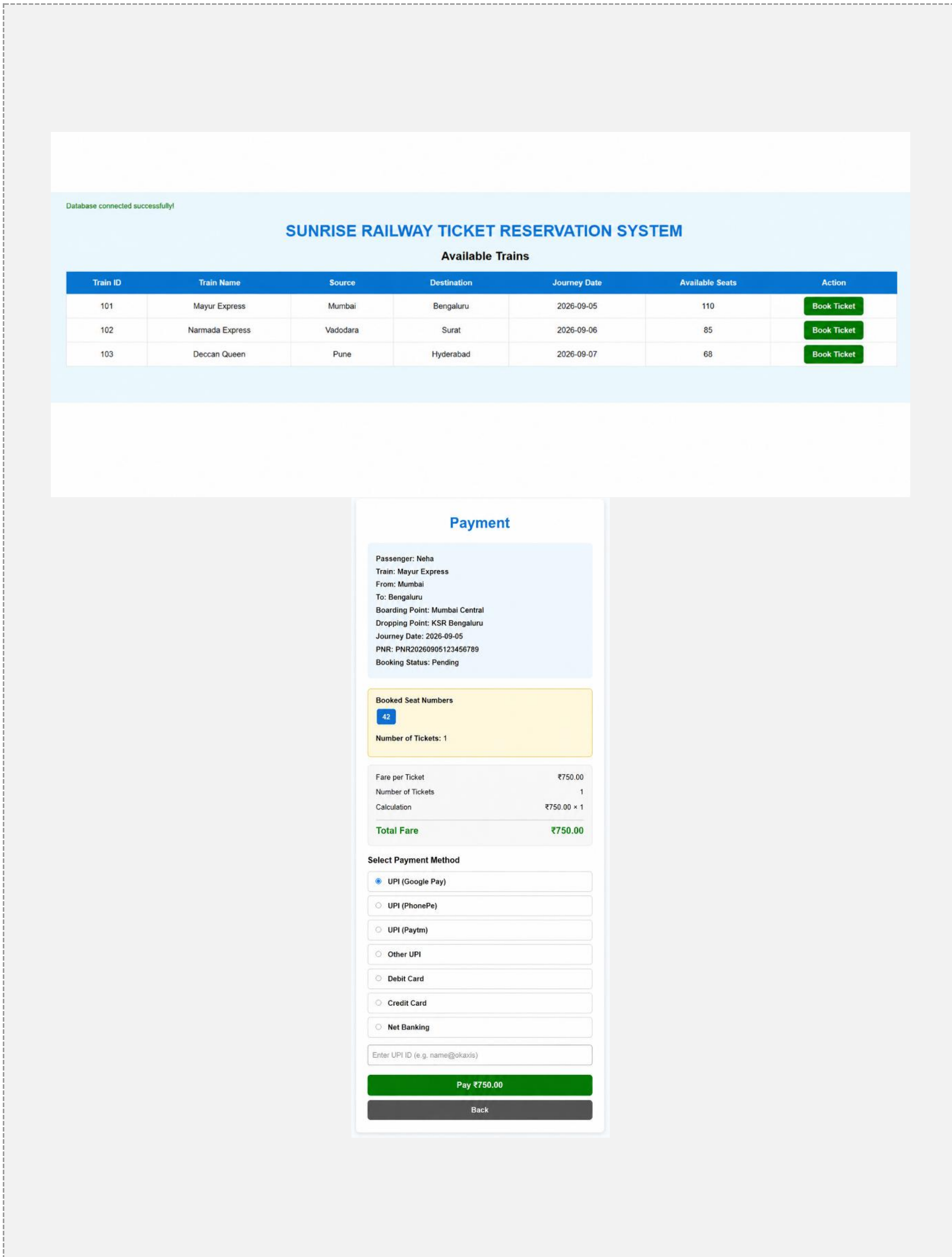
On booking, a unique Ticket ID is generated and displayed, and the record is inserted into the passengers table.

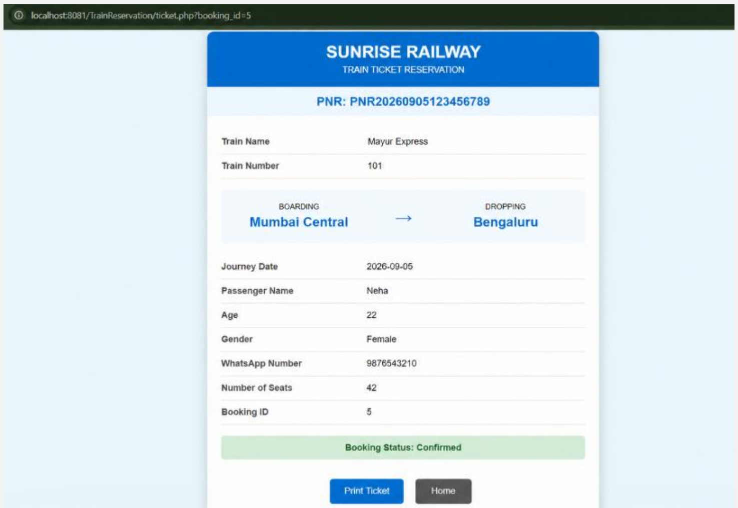
On viewing a ticket by its ID, all stored details (name, age, gender, source, destination, date, class) are displayed correctly.

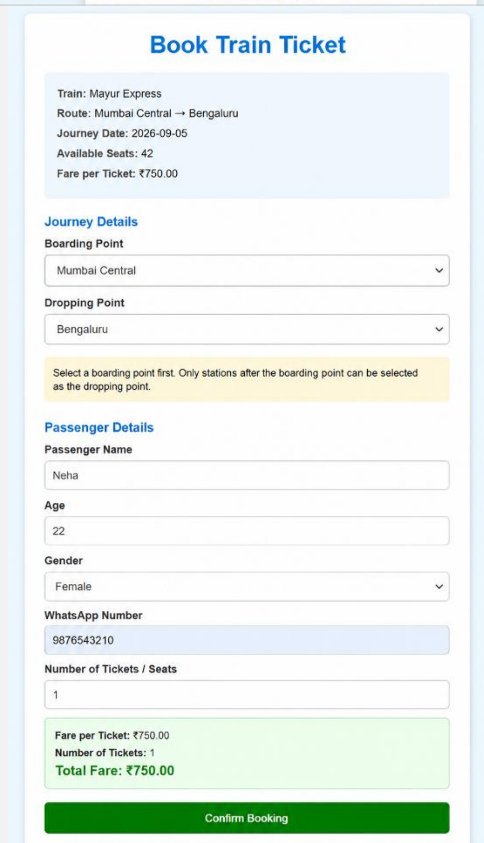
On cancellation, the corresponding record is deleted from the database and no longer appears in subsequent queries.

Invalid or non-existent ticket IDs are handled gracefully with an appropriate message.

OUTPUT SCREENSHOT:







RESULT: Thus an online train ticket reservation system was designed and implemented, and booking, viewing and cancellation of tickets were verified using the database.

Experiment No: 24

ONLINE COLLEGE ADMISSION FORM

AIM

To create an online college admission form using Python (Tkinter) and MySQL that accepts candidate details and generates a unique admission number for each candidate.

PROCEDURE

- Ensure the latest version of Python and MySQL are installed and the MySQL service is running.
- Install the required modules (tkinter, mysql-connector-python) using pip.
- Create the admission database and the students table to store candidate details.
- Establish a connection between the Python program and the MySQL database.
- Design the Tkinter form with input fields for first name, last name, age, gender, community, state and marks.
- On clicking Submit, generate a unique admission number and insert the candidate's details into the students table.
- Display the generated admission number to the user in a message box.
- Run the program and verify the record is stored correctly in the database.

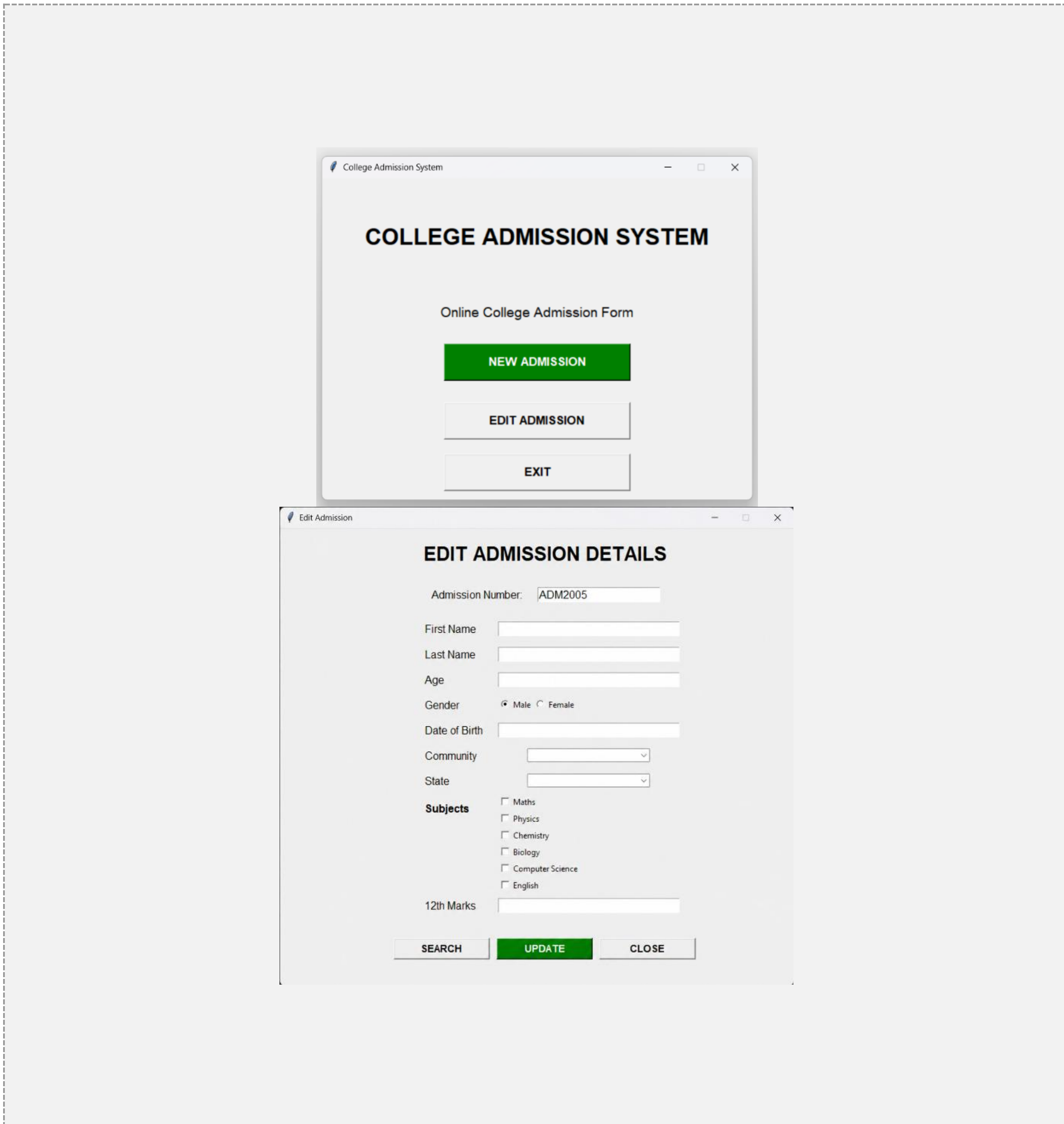
EXPECTED OUTPUT

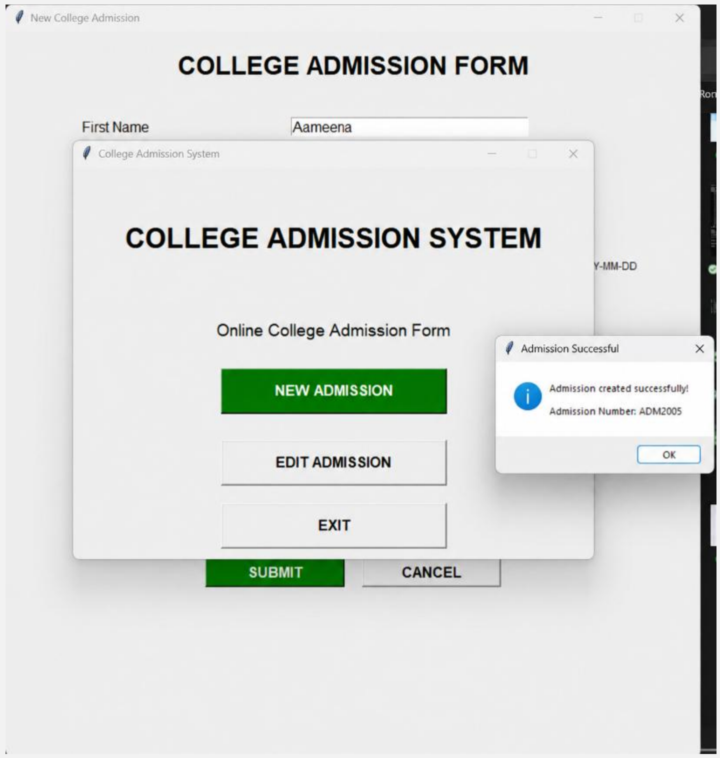
The admission form window opens with all input fields visible and editable.

On submitting valid details, a unique admission number is generated and shown in a confirmation dialog.

The candidate's details, along with the generated admission number, are correctly stored as a new row in the students table.

OUTPUT SCREENSHOT:





RESULT: Thus an online college admission form was created, and a unique admission number was successfully generated and stored for each candidate.

Experiment No: 25

QR-ENABLED AUTOMATIC BUS TICKET BOOKING SYSTEM

AIM

To design a bus ticket booking system using Python (Tkinter) and MySQL that books a ticket for a passenger and generates a QR code containing the ticket details as a digital ticket.

PROCEDURE

- Verify the installed versions of Python and MySQL using "python --version" and "mysql --version".
- Install the required modules: tkinter, mysql-connector-python, qrcode and pillow.
- Create the bus_db database and the passengers table to store booking details.
- Establish a connection between the Python program and the MySQL database using mysql.connector.
- Design the Tkinter interface with fields for name, age, source and destination.
- On clicking Book Ticket, generate a unique ticket ID, insert the booking into the database, and encode the ticket details into a QR code using the qrcode module.
- Display the generated QR code image within the Tkinter window using PIL/ImageTk.
- Run the program and verify the booking is stored and the QR code is generated correctly.

EXPECTED OUTPUT

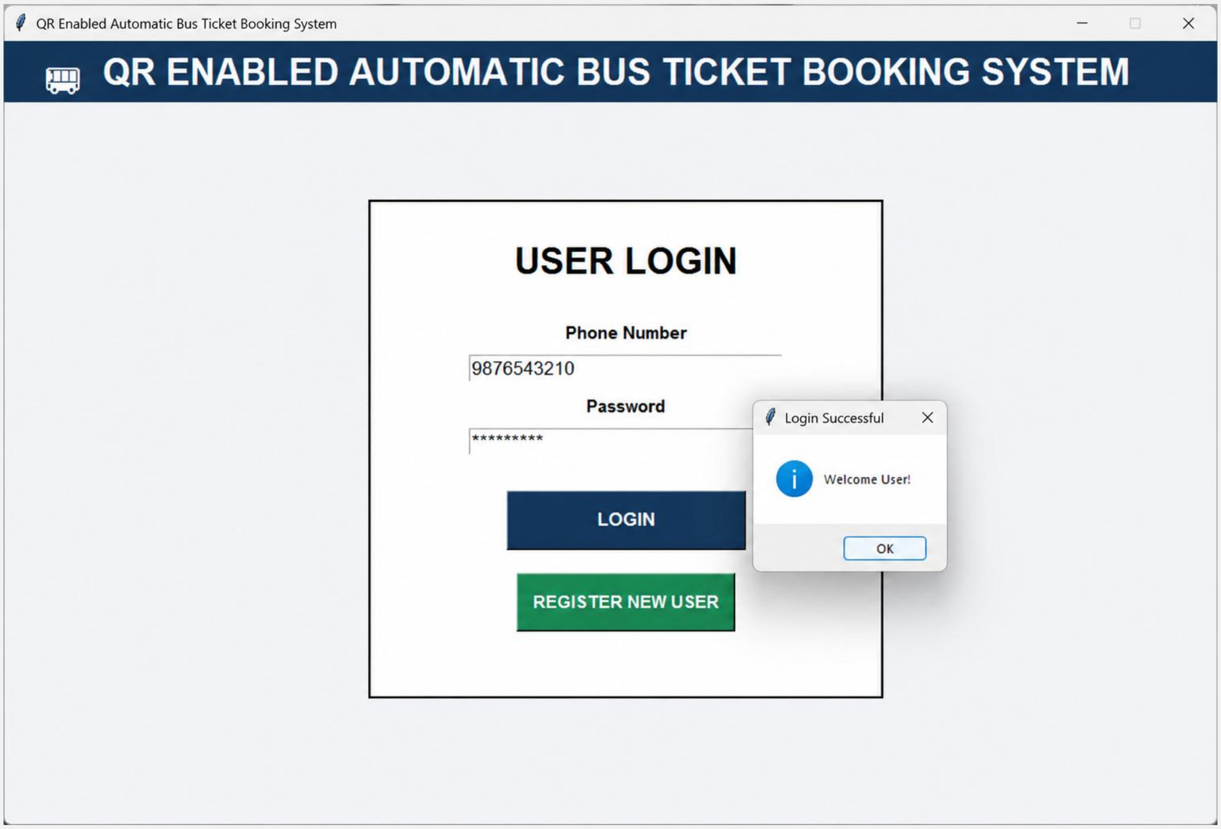
The booking window opens with fields for name, age, source and destination.

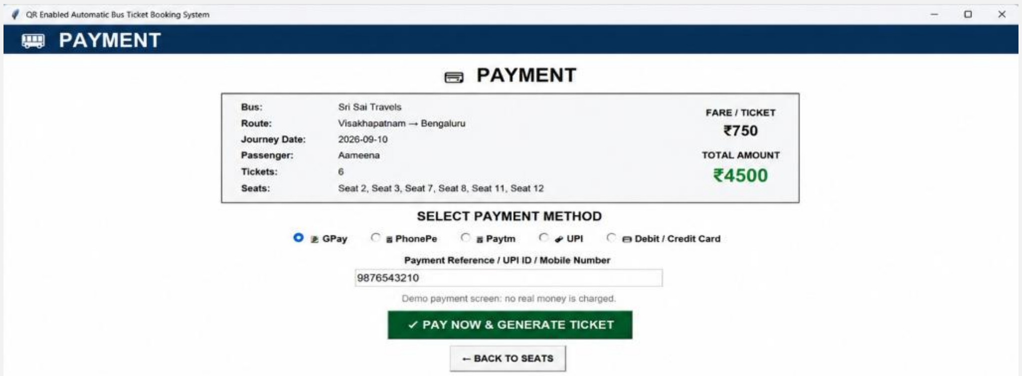
On booking, a unique ticket ID is generated and the passenger's details are inserted into the passengers table.

A QR code image is generated and displayed in the window, encoding the ticket ID, name and route (source → destination).

Scanning the QR code (with any QR reader) reveals the encoded ticket information.

OUTPUT SCREENSHOT:





RESULT: Thus a QR-enabled bus ticket booking system was implemented, successfully booking a ticket and generating a corresponding QR code as a digital ticket.